Software Requirements Specification

for

furlink

Version 1.0

Prepared by

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Contents

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft Type and Number	Full Name	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

1.1 Document Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define and describe all functional and non-functional requirements for the furlink, version 1.0. This serves as a detailed guide for stakeholders including developers, project managers, testers, and end-users. Ensuring that all sides share a common understanding of the system's goals, constraints, and expected behavior.

1.2 Product Scope

This proposed project aims to develop a website platform to connect dog or cat owners with dog or cat grooming service providers in Makati City, creating more efficient access to pet grooming services. Meanwhile, it also allows service providers to have a wider and a more inclusive reach to their potential customers and organization of their booking system. As a result, this proposed project aims to increase the economic growth through amplifying the pet service industry, making this proposal aligned with SDG number 8.

The project includes dog or cat owners in Makati City that seeks new service options and are open to digital solutions. Another inclusion are small to medium-sized dog or cat grooming service providers based in Makati City that are not located in well-known areas

1.3 Intended Audience and Document Overview

It's essential to acknowledge the target beneficiaries of this project, especially people who are involved in the pet care industry. More specifically, this project is intended to the following groups of people:

- Pet Owners – This proposed project helps pet owners to have efficient access to reliable and nearby pet service providers. Improve user experience and provide convenience in booking pet services through the online platform.
- Pet Service Providers – This proposed project helps pet service providers reach a wider market than their usual local area. Increase visibility and improve organization of booking and scheduling through the system's digitalization.
- Local Communities – This proposed project helps the local community in making quality pet services accessible, improving their standards of pet care in the area, and boosting pet related services through gaining business activity on the platform.

1.4 Definitions, Acronyms and Abbreviations

AI (Artificial Intelligence) – Technology that enables machines to perform tasks that typically require human intelligence, such as image generation or prediction.
API (Application Programming Interface) – A set of tools and protocols that allow different software components to communicate.

Client – The end user or organization for whom the system is being developed.
Cloud-Based Database – A database hosted on remote servers accessed through the internet rather than using local or on-premise infrastructure.

Dashboard – A user interface that provides analytics, summaries, and performance indicators related to system activities.
Deliverables – Documents, modules, or outputs required for submission according to the PBL schedule.
Database – A structured collection of data stored and accessed electronically, in this project handled through Supabase.

furlink – The name of the proposed web-based pet grooming service booking platform.

Grooming Preview (AI Grooming Tool) – A feature that generates an estimated visual representation of a pet's haircut using AI technology.

HTML (HyperText Markup Language) – Standard language used to structure and display content on the web.

Local Service Provider – A small or medium-sized pet grooming business within Makati City using the platform to reach customers.
Messaging Channel – An in-platform communication feature that allows pet owners and service providers to exchange messages.
MVP (Minimum Viable Product) – An early version of the system that includes essential features for initial adoption and testing.

PBL (Project-Based Learning) – The instructional approach governing the project timeline, milestones, and deliverables.
Pet Owner – The end user looking to book grooming services for their dog or cat.
Platform – The Furlink website where users interact, book, and manage grooming services.

SDG (Sustainable Development Goal) – UN-defined global goals; the project aligns with SDG No. 8, focusing on economic growth.
Service Listing – A feature where grooming providers publish available services and pricing.
Service Provider / Pet Service Provider – A business offering dog or cat grooming services through the platform.
Supabase – A cloud-based backend platform providing database, authentication, and storage services.

UI (User Interface) – The visual elements users interact with in the platform.
UML (Unified Modeling Language) – A standardized modeling language used to represent software structure, behavior, and architecture in diagrams.
User – Any individual interacting with the system, including pet owners and service providers.
User Registration – The account creation process required before accessing platform features.

Web Platform – The online system where Furlink's services are delivered and accessed via a browser.

1.5 Document Conventions

<In general this document follows the IEEE formatting requirements. Use Arial font size 11, or 12 throughout the document for text. Use italics for comments. Document text should be single spaced and maintain the 1" margins found in this template. For Section and Subsection titles please follow the template.

TO DO: Describe any standards or typographical conventions that were followed when writing this SRS, such as fonts or highlighting that have special significance. Sometimes, it is useful to divide this section to several sections, e.g., Formatting Conventions, Naming Conventions, etc.>

1.6 References and Acknowledgments

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2 Overall Description

2.1 Product Overview

As a startup, furlink aims to modernize the way pet grooming services are booked and managed in the Philippines. Unlike platforms such as Agoda or Airbnb—which focus on property and hotel bookings—furlink specializes in connecting dog or cat owners with pet grooming service providers through a streamlined digital platform.

furlink offers a hassle-free experience for both customers and service providers. It enables grooming businesses to advertise their services while allowing pet owners to discover, schedule, and manage appointments in one place. Additionally, the platform integrates an innovative AI-powered grooming preview tool, giving users a visual reference of potential pet haircut styles before booking—bridging the gap between customer expectations and actual grooming outcomes.

Designed as a local-first startup solution, furlink initially targets grooming services within select cities, with plans to expand as adoption grows.

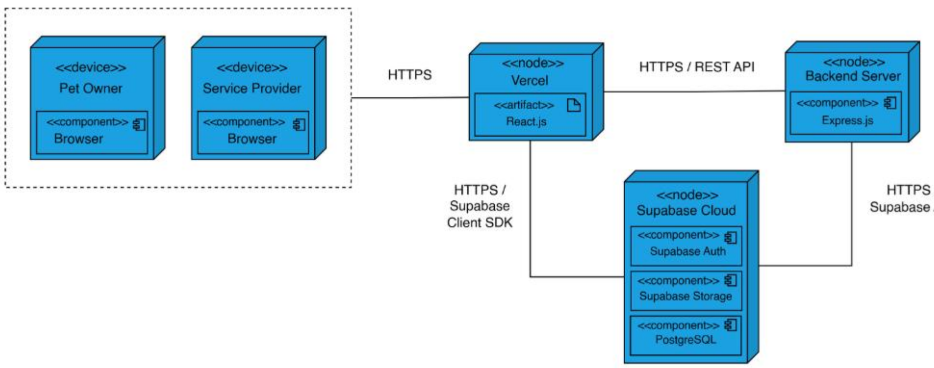

Figure 1 furlink Deployment Diagram

2.2 Product Functionality

This project will include the following features:

1. User registration and login: Allows both dog or cat owners and service providers to create an account and login to their account.
2. Service listing: Allows service providers to post their services on the platform.
3. Booking: Enables dog or cat owners to book pet services on the platform.
4. AI pet haircut generator: Enables dog or cat owners to upload a photo of their pet and receive an AI-generated preview for their desired haircut style.
5. Pet service provider recommendation: Provides dog or cat owners recommended pet service providers based on their activities.
6. Dashboard: Allows pet service providers to track bookings, revenue, and customer insights.
7. Feedback form: Allows dog or cat owners to submit their review to the service provider.

2.3 Design and Implementation Constraints

The development of furlink system is put through several constraints that will limit the options that are available to develop and design furlink. First, the team must follow the PBL-mandated schedule for the submission of all project deliverables which gives a limited time for development. This limited development timeframe affects the level of system refinement, iteration cycles, and the complexity of features that can be implemented within the project period. Second, the system will use Supabase a cloud-based backend database with no support for any on-premises servers. This requires for the system to solely rely on Supabase’s authentication, storage, and other database services.

This project will exclude the following features:

1. Booking of pet care services with insurance.
2. Other pet services such as veterinary care, pet supplies, etc.
3. Pet transportation services
4. The system does not process any payments. It does not use in-app payment through third-party gateways.
5. In-depth financial reporting tools.

2.4 Assumptions and Dependencies

Assumptions

- If there is a potential client that does not have a full working system.
- If there is a potential client, they will fully fund the financial requirements of the project.
- There will be one pet grooming service provider that will be onboard.

Constraints

- Limited financial resources.
- Heavily relies on open-source and free-tier tools.
- Limited time is given.
- The MVP aims to be completed by February 21, 2026.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users. For your project, you only need to be concerned with the main thermostat (not the mobile app) and can use the graphic from the project description as the basis for your user interface..
TO DO: Provide the graphic for the thermostat user interface and provide a basic description as to how users will interact (e.g. tough screen, menus, etc.).>

3.1.2 Hardware Interfaces

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~~<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware. You are not required to specify what protocols you will be using to communicate with the hardware, but it will be usually included in this part as well.~~
~~TO DO: Please provide a short description of the different hardware interfaces. This can simply be a list of the devices you must interact with at this point. For sensors, you can assume that each has a basic "read" interface to get the current values. Each sensor in this system uses English units.>~~

The furlink web app requires appropriate hardware and network configurations to ensure reliable and efficient system performance. The system operates using standard HTTP/HTTPS protocols to facilitate secure and seamless communication between user devices and backend services, maintaining compliance with modern web standards and data protection practices.

furlink is hosted on a cloud-based infrastructure, where core application files such as HTML, JavaScript, and media assets are stored and delivered through cloud hosting services. While no specialized hardware is required on the user side, sufficient device storage is recommended to support temporary caching of data for improved responsiveness. In environments protected by network firewalls, the necessary ports must be properly configured to enable access to the platform's cloud-hosted services, ensuring uninterrupted connectivity between users and the furlink system.

3.1.3 Software Interfaces

<*Describe the connections between this product and other specific software components (in your case, just the mobile app that can send commands).*>

3.2 Functional Requirements

< *Functional requirements capture the intended behavior of the system. This behavior may be expressed as services, tasks or functions the system is required to perform. This section is the direct continuation of section 2.2 where you have specified the general functional requirements. Here, you should list in detail the different product functions.*

3.2.1 F1: The system shall …

3.2.2 <Functional Requirement or Feature #2>

…

3.3 Use Case Model

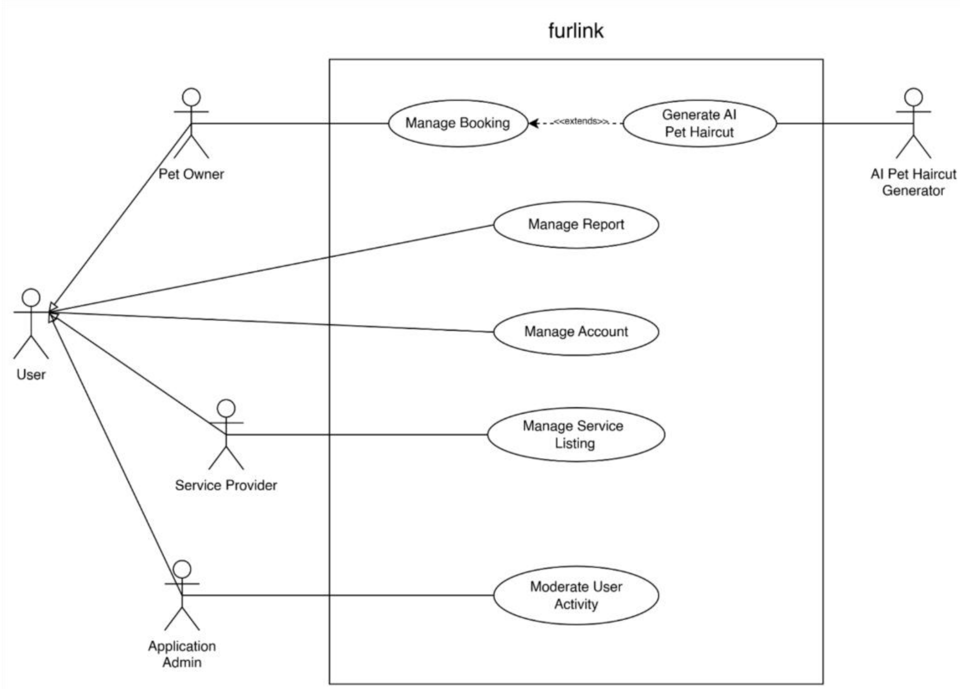

Figure 2 Use Case Diagram

3.3.1 Use Case #1 (use case name and unique identifier – e.g. U1)

TO DO: Provide a specification for each use case diagram

Author – Identify team member who wrote this use case

Purpose - What is the basic objective of the use-case. What is it trying to achieve?

Requirements Traceability – Identify all requirements traced to this use case

Priority - What is the priority. Low, Medium, High. Importance of this use case being completed and functioning properly when system is depolyed

Preconditions - Any condition that must be satisfied before the use case begins

Post conditions - The conditions that will be satisfied after the use case successfully completes

Actors – Actors (human, system, devices, etc.) that trigger the use case to execute or provide input to the use case

Extends – If this is an extension use case, identify which use case(s) it extends

Flow of Events

1. Basic Flow - flow of events normally executed in the use-case
2. Alternative Flow - a secondary flow of events due to infrequent conditions
3. Exceptions - Exceptions that may happen during the execution of the use case

Includes (other use case IDs)

Notes/Issues - Any relevant notes or issues that need to be resolved

3.3.2 Use Case #2

…

4 Other Non-functional Requirements

4.1 Performance Requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.
TODO: Provide performance requirements based on the information you collected from the client/professor. For example, you can say "P1. The secondary heater will be engaged if the desired temperature is not reached within 10 seconds">

4.2 Safety and Security Requirements

<*Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied. Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements.*
TODO:

- *Provide safety/security requirements based on your interview with the client - again you may need to be somewhat creative here. At the least, you should have some security for the mobile connection.*

4.3 Software Quality Attributes

<*Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.*

*TODO: Use subsections (e.g., 4.3.1 Reliability, 4.3.2 Adaptability, etc…) provide requirements related to the different software quality attributes. Base the information you include in these subsections on the material you have learned in the class. Make sure, that you do not just write "This software shall be maintainable…" Indicate how you plan to achieve it, & etc…Do not forget to include such attributes as the design for change (e.g. adapting for different sensors and heating/AC units, etc.). Please note that you need to include **at least** 2 quality attributes. You can Google for examples that may pertain to your system.>*

5 Other Requirements

<*This section is **<u>Optional</u>**. Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.*>

Appendix A – Data Dictionary/ERD

<Data dictionary is used to track all the different variables, states and functional requirements that you described in your document. Make sure to include the complete list of all constants, state variables (and their possible states), inputs and outputs in a table. In the table, include the description of these items as well as all related operations and requirements.>

Appendix B - Group Log

<Please include here all the minutes from your group meetings, your group activities, and any other relevant information that will assist in determining the effort put forth to produce this document>

Appendix C – Test Plan/Test Cases